Thrombophilia: Antiphospholipid Syndrome

Objective:

To outline the main clinical and laboratory features of the antiphospholipid syndrome (APS) and to describe its anticoagulant management.

Background:

APS is an acquired hypercoagulable state characterized by the persistent presence of autoantibodies against proteins bound to cell membrane phospholipids. It is associated with thrombosis (venous, arterial, or microvascular) and/or pregnancy complications such as recurrent miscarriage, late pregnancy loss, or pre-eclampsia. There may be accompanying features such as thrombocytopenia, livedoid rash, cardiac valvulopathy, renal disease and neurologic symptoms. APS may occur in the setting of underlying autoimmune disease such as systemic lupus erythematosus (**secondary APS**) or may occur in isolation (**primary APS**). The term “obstetric APS” is used to emphasize the pregnancy-related complications associated with antiphospholipid syndrome, such as recurrent miscarriage, fetal loss, or severe preeclampsia. However, this designation does not imply that these patients are free from other manifestations of the disease. In fact, individuals classified as having obstetric APS may also experience macrovascular complications (such as deep vein thrombosis or stroke) or microvascular complications (such as small-vessel thrombosis), either before, during, or after pregnancy. This document focuses on non-obstetrical manifestations of APS.

Diagnosis:

The diagnosis of APS should be made carefully and in consultation with a specialist because of the potential for both false positive and false negative laboratory tests. In addition, a diagnosis of APS has important treatment implications because such patients may require long-term anticoagulant therapy. APS is diagnosed based on expert consensus criteria (Sydney criteria) and requires the presence of at least one laboratory **and** one clinical criterion.

APS classification was updated based on new expert consensus criteria in 2023 which are more reflective of current clinical understanding of this disease (**Figure 1**).The 2023 classification criteria for antiphospholipid syndrome (APS) require an initial positive entry criterion, followed by weighted additive criteria organized into two laboratory domains and six clinical domains. To classify a patient as having APS, there must be at least 3 points from clinical domains and at least 3 points from laboratory domains.

When validated in two international cohorts of patients with suspected APS, these new criteria demonstrated higher specificity compared to the 2006 revised Sapporo classification. However, this improvement in specificity has come at the cost of reduced sensitivity. As a result, some patients who previously met the 2006 criteria no longer meet the 2023 criteria, even though they may still have clinically significant disease. It is important to note that the classification criteria have been designed for research purposes and the reduction in sensitivity may lead to underdiagnosis in the clinical setting. As such, the British Society of Hematology 2024 APS guidelines suggest not using the 2023 classification criteria routinely in the clinical setting. Instead, they suggest that diagnosis of APS still be made with a combination of at least one persistent antiphospholipid antibody (minimum 12 weeks apart in one of lupus anticoagulant, anticardiolipin IgG or IgM, anti- β2GPI IgG or IgM) with either unprovoked/weakly provoked thrombosis, or pregnancy morbidity.

Figure 1: 2023 ACR/EULAR classification criteria

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Please refer to the original article for full details of each footnote in figure 1. Some selected definitions from the figure are listed below:

- Moderate positive aCL/aβ2GPI = 40-79 Units, based on standardized enzyme-linked immunosorbent assay (ELISA) results
- High positive aCL/aβ2GPI = 80 Units or more, based on standardized enzyme-linked immunosorbent assay (ELISA) results
- ELISA results are not equivalent to other methods of detection such as chemiluminescent immunoassay which may be in wider use in the clinical setting
- VTE with a high-risk profile = 1 or more major OR 2 or more minor VTE risk factors
  - Major VTE risk factors include any of: active malignancy, hospital admission for at least 3 days and within 3 months, major trauma within 1 month, or surgery for >30 minutes within 3 months prior to the VTE event
  - Minor VTE risk factors include any of: active systemic autoimmune disease, active inflammatory bowel disease, acute/active severe infection, central venous catheter in the same vascular bed, hormone replacement therapy, estrogen-containing oral contraceptives, ongoing in vitro fertilization treatment, travel ≥8 hours, BMI ≥30kg/m2, pregnancy or within 6 weeks postpartum, prolonged immobilization, or surgery for <30 minutes within 3 months prior to the VTE event
- Arterial thrombosis with a high-risk profile = 1 or more high risk factors or 3 or more moderate risk factors
  - High risk factors include any of: arterial hypertension (systolic BP ≥180 mm Hg or diastolic BP ≥110 mm Hg), estimated glomerular filtration rate ≤60 mL/min for more than 3 months, diabetes mellitus with organ damage or long duration (type 1 for ≥20 years or type 2 for ≥10 years), or hyperlipidemia with total cholesterol ≥8 mmol/L or LDL >4.9 mmol/L
  - Moderate risk factors include any of: arterial hypertension on treatment, current tobacco smoking, diabetes mellitus with no organ damage or short disease duration, hyperlipidemia on treatment or with total cholesterol above normal range and <8 mmol/L or LDL above normal range and <4.9 mmol/L, or BMI ≥30 kg/m2

Antithrombotic Therapy of APS:

Due to the complexity and potential severity of APS, diagnosis, and treatment of patients with APS should be undertaken in consultation with a specialist.

High risk APS: Risk stratification of APS may inform antithrombotic selection. Various definitions exist but the most widely accepted high risk characteristics are ‘triple positivity’ (positive for LA, aCL and aß2GP1) or a history of confirmed arterial thrombosis. Some also consider a history of small vessel thrombosis or organ involvement, heart valve disease according to criteria, or recurrent thrombosis as high-risk characteristics.

Acute venous thrombosis: LMWH is preferred over unfractionated heparin (UFH) in patients with venous thrombosis whose baseline aPTT is prolonged by a LA because of the difficulties with monitoring UFH in this situation. Patients with high-risk APS should be treated with warfarin administered to a target INR of 2.0 to 3.0 (see “DOACs in APS” below). Whether DOACs can be used in low-risk patients is unknown as there are no robust randomized control trials and conflicting results from observational studies.

Acute arterial thrombosis: The treatment of patients with an acute arterial thrombosis is controversial as data are limited. However, most guidelines currently recommend anticoagulation therapy with warfarin a

target INR of 2.0 to 3.0.

An older study assessed secondary ischemic stroke prevention with acetylsalicylic acid (ASA) warfarin (INR target of 1.4-2.8). A subset of this study included patients with low titer antiphospholipid antibodies that were only measured once, rendering the diagnosis of APS uncertain; in this subset ASA and warfarin were equivalently effective. Thus, antiplatelet therapy or anticoagulant therapy are seen as options in this setting but anticoagulant therapy with warfarin administered to a target INR of 2.0 to 3.0 is generally preferred.

DOACs in APS: Randomized control trials have shown that APS patients treated with rivaroxaban or apixaban experience more strokes and myocardial infarctions than patients treated with warfarin administered to a target INR of 2.0 to 3.0. These trials generally enrolled a high proportion of triple positive patients. Therefore, warfarin is currently the preferred long-term anticoagulant for patients with high-risk APS. Patients who decline warfarin or who cannot tolerate warfarin should be informed about the available evidence, possible reduced benefit of DOACs relative to warfarin, and referred to a hematologist or thrombosis specialist. There is uncertainty about the safety of DOAC in low-risk patients and specialist consultation is recommended.

Long-term anticoagulant management: Since the risk of recurrent thrombosis in patients with APS is high, long-term anticoagulation is usually required. Consultation with a specialist is recommended for patients with APS. All patients with APS should have aggressive reduction of their modifiable cardiovascular risk factors. Patients who also have systemic lupus erythematosus may benefit from the addition of hydroxychloroquine or other lupus-specific therapies; specialist consultation is strongly advised. It is not known whether anticoagulation may be stopped safely if the laboratory criteria for APS are no longer present on later follow-up.

Recurrent thrombosis: There is limited data on patients who develop objectively confirmed recurrent thrombosis - treatment options include switching to warfarin with a target INR of 2.0 to 3.0 (if previously on a DOAC), higher-intensity warfarin (target INR 3.0 to 4.0), therapeutic dose LMWH, or the combination of low dose ASA and conventional warfarin therapy. Specialist consultation is recommended for these patients. In all cases the suspected recurrence should be objectively confirmed, and the adequacy of anticoagulation assessed before an event is labelled a recurrence, given the long-term implications of failure of therapeutic anticoagulation and the high frequency of non-compliance, particularly with DOACs and LMWH which cannot be easily measured.

Laboratory monitoring of anticoagulant therapy: Some patients with APS have a prolongation in the INR before anticoagulant therapy is commenced. In such patients, alternate monitoring approaches may be necessary. Consultation with an expert is strongly recommended to optimize therapy. Point of care INR determinations appear to be particularly prone to error in patients with APS. Periodic reconfirmation of the reliability of test results is required for patients being monitored with point of care devices – this is performed by calibrating the point of care result with a result obtained through conventional testing through venipuncture.

Special Considerations:

Asymptomatic (without thrombosis) patients with positive antiphospholipid test results: Due to the widespread use of the aPTT in clinical practice, a LA may be detected in otherwise asymptomatic patients who do not have the clinical criteria for APS. A detailed clinical history should be taken to exclude thrombotic events that were previously missed. Although asymptomatic patients with both LA and other APS markers may be at increased risk for thrombotic complications, there is no consensus on the role of primary antithrombotic prophylaxis or low-dose ASA. Nonetheless, some experts consider the use of low-dose ASA for primary prevention in asymptomatic carriers of APS markers with a high-risk antibody profile, particularly those with SLE or additional thrombotic risk factors. All asymptomatic patients with positive antiphospholipid tests should receive aggressive thromboprophylaxis in high-risk situations. APS may be associated with other manifestations such as cardiac valvulopathy and livedoid rash; use of anticoagulants in these settings remains unclear.

Pregnant women with antiphospholipid antibodies: It is recommended that pregnant wom
for obstetric APS without a history of venous/arterial thrombosis receive prophylactic-dose LM
with low-dose ASA for the duration of their pregnancy; however, it is important to note that the
such management has not been validated in well-designed clinical trials. The role of prophylactic-dose LMWH/UFH and low-dose ASA in women with persistent antiphospholipid antibodies and a single late pregnancy loss has not been well studied. Low-dose ASA is often used in pregnant women with persistent antiphospholipid antibodies to reduce the risk of pre-eclampsia. Post-partum prophylaxis is also frequently used in these patients even in the absence of a prior history of thrombosis, although this practice is not evidence-based.

Catastrophic antiphospholipid antibody syndrome: Catastrophic antiphospholipid antibody syndrome (CAPS) is a rare clinical manifestation presenting with a fulminant onset of multiorgan microvascular thrombosis, oftentimes in patients without a prior history of APS. Management consists of aggressive anticoagulation, plasmapheresis, and immunosuppression. Despite treatment, there is a high rate of long-term morbidity and mortality. Diagnosis requires identification of one or more antiphospholipid antibodies but does not require tissue biopsy evidence of microvascular thrombosis in a compatible clinical setting. Management should occur in expert centers under the direction of experienced clinicians. Anti-complement therapy may be of benefit in selected patients and should be considered for critically ill patients and those not responding to “standard” therapies.

Pediatrics: For children with venous thromboembolism (VTE) in the setting of antiphospholipid antibodies, anticoagulation as per general recommendations for VTE management in children is recommended. Pediatricians with expertise in thromboembolism should manage, where possible, pediatric patients with thromboembolism. When this is not possible, a combination of a neonatologist/pediatrician and a pediatric hematologist or an adult hematologist, supported by consultation with an experienced pediatric hematologist, is recommended.

Other Relevant

- Deep Vein Thrombosis (DVT): Treatment
- Pregnancy: Thromboprophylaxis
- Pregnancy: Venous Thromboembolism Treatment
- Pulmonary Embolism (PE): Treatment
- Unfractionated Heparin, Low Molecular Weight Heparin, and Fondaparinux
- Warfarin

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